Nitu Alexandru-Nicolae, 314CD - Partea I

For my project, I've decided to use the OpenWeatherMap API so I could record various data about the weather. The way I've built my dataframes is:

-first and foremost, I got myself an API key from their website

-afterwards, I chose 18 random capital cities so I could get enough instances of data (720), which are then split into 510 and 210 for the training and testing sets respectively

-before building the main table itself, I chose only 8 attributes that interested me the most. those are: Date & Time, Temperature, Weather, Wind Speed, Humidity, Feels Like, Cloud Coverage %, and City

-once I built the main table, I split it randomly into the training and testing sets like I've previously mentioned and turned the two tables into .csv files

Now, for the data analysis part:

-for the missing values, I calculated the number of values and percentages themselves (which happen to be 0 due to the API itself not having NA data) and erased all of the instances of data containing missing values

-here are some basic stats for both sets, first the training set and then the testing set:

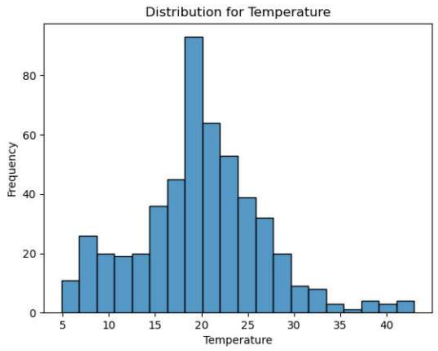
	Temperature	Wind Speed	Humidity	Feels Like	Cloud Coverage %
count	510.000000	510.000000	510.000000	510.000000	510.000000
mean	19.864804	3.349608	64.215686	19.416549	48.601961
std	6.838656	1.968490	19.646096	7.228838	40.786285
min	4.880000	0.260000	14.000000	0.900000	0.000000
25%	15.920000	1.920000	50.000000	15.502500	4.250000
50%	19.660000	2.955000	67.000000	19.665000	46.000000
75%	23.655000	4.312500	80.750000	23.530000	97.000000
max	42.960000	11.340000	100.000000	41.330000	100.000000

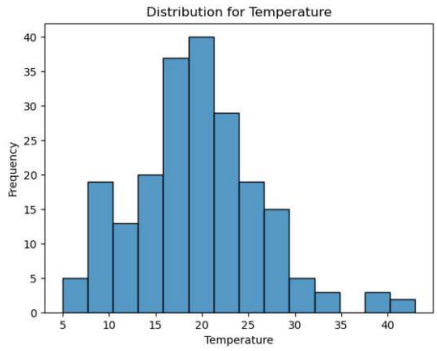
	Date & Time	Weather	City
count	510	510	510
unique	40	3	18
top	2026-05-26 09:00:00	Clouds	Ottawa
freq	17	253	34

	Temperature	Wind Speed	Humidity	Feels Like	Cloud Coverage %
count	210.000000	210.000000	210.000000	210.000000	210.000000
mean	19.341905	3.094667	63.823810	18.807143	47.152381
std	6.869846	1.918494	18.726498	7.391543	41.186304
min	4.960000	0.140000	14.000000	0.680000	0.000000
25%	14.807500	1.457500	50.000000	14.017500	5.000000
50%	19.130000	2.825000	66.500000	19.010000	43.000000
75%	23.437500	4.432500	78.000000	23.257500	93.000000
max	42.960000	9.720000	99.000000	41.330000	100.000000

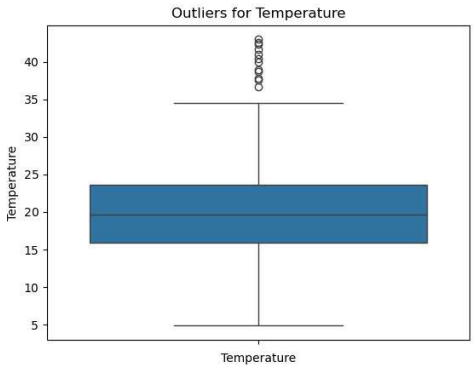
	Date & Time	Weather	City
count	210	210	210
unique	40	3	18
top	2026-05-30 03:00:00	Clouds	Ankara
freq	9	92	18

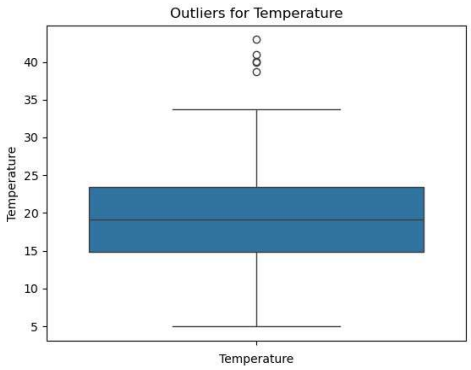
-I've also made plots for every characteristic in both the training and testing sets, using histograms for numerical values and countplots for the rest. Going forward, every picture on the left will be data representations for the training set, and every picture on the right will be for data from the testing set:



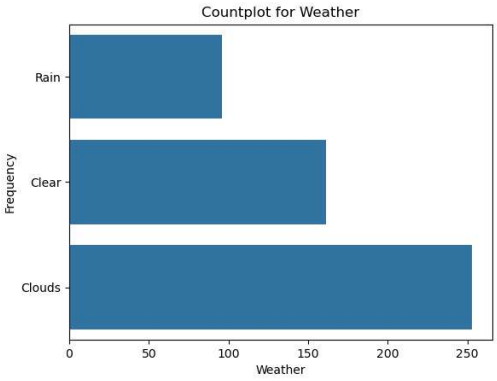


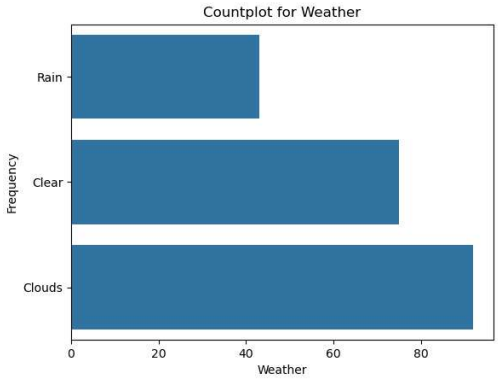
As we can see, both sets don't really have values between 35-40 degrees C, which we'll have to keep in mind for later; a majority of the current temperatures are around 20 degrees, which checks out due to it being late spring; here's the boxplot as well:

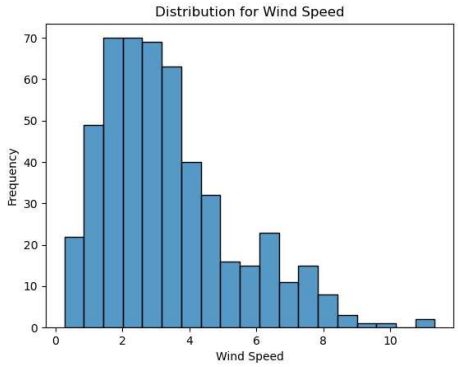


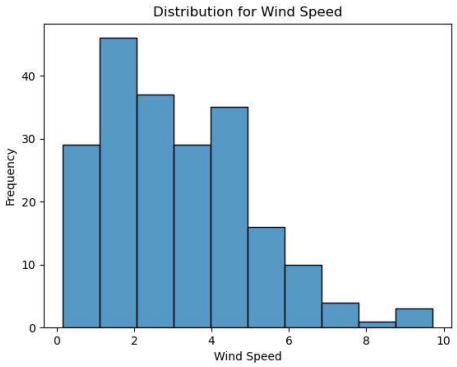


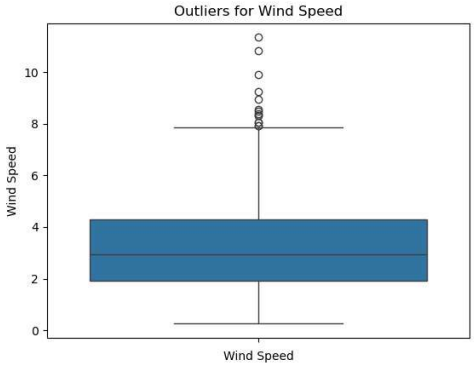
In both cases, the outliers are values over approximately 35 degrees C; this is because any temperature above that is considered extreme during this period of time.

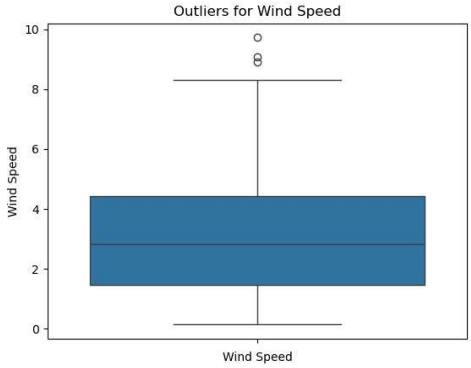




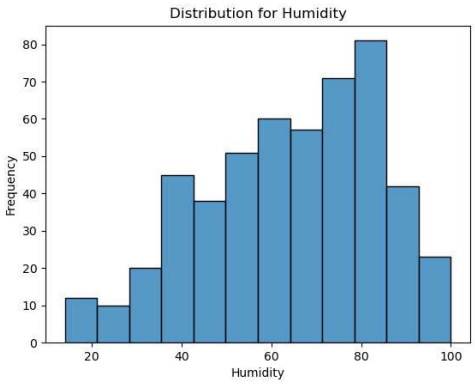


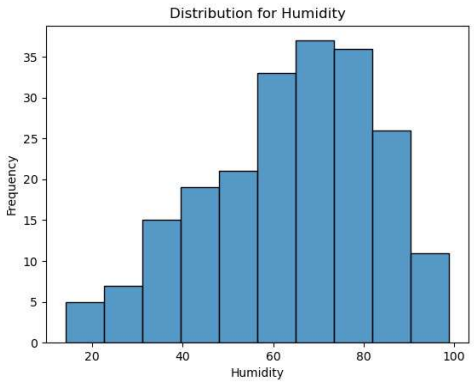




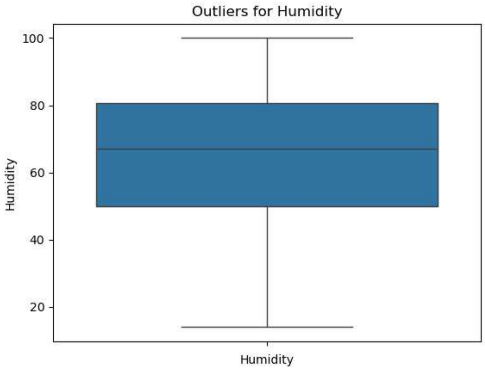


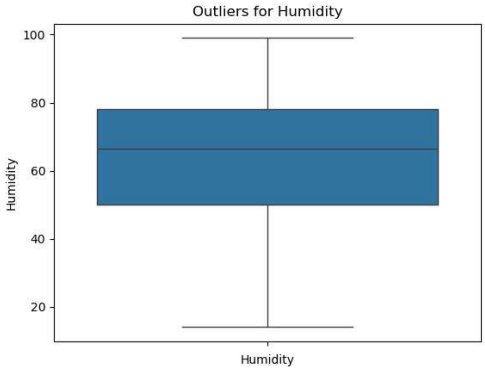
In both cases, the outliers for wind speeds are above 8 km/h

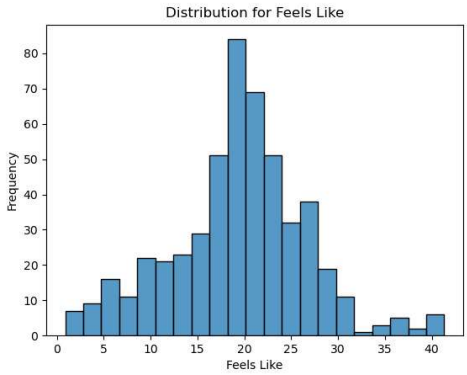


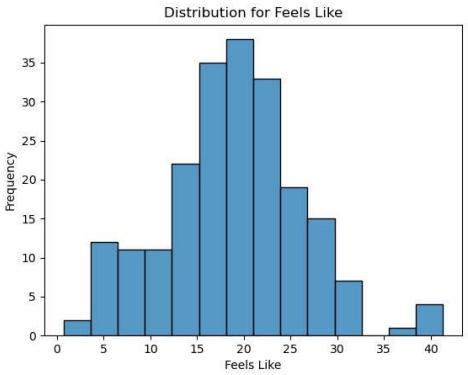


Both distributions are fairly similar, therefore the boxplots will be near identical

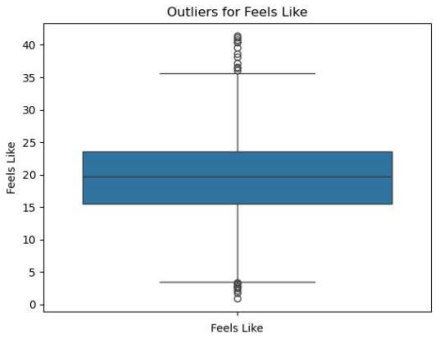


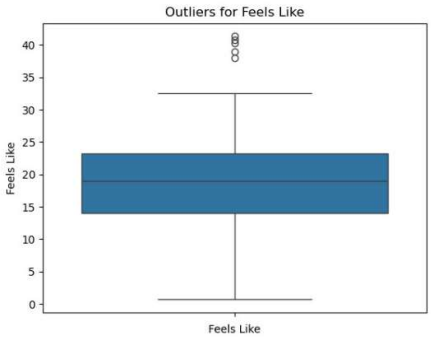


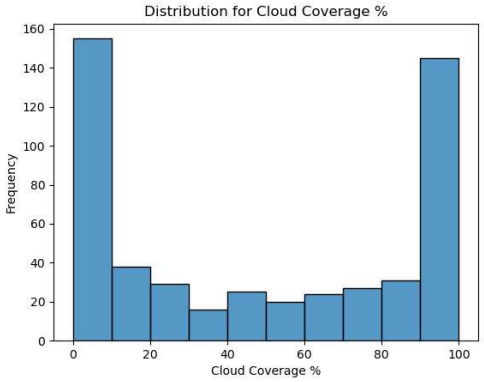


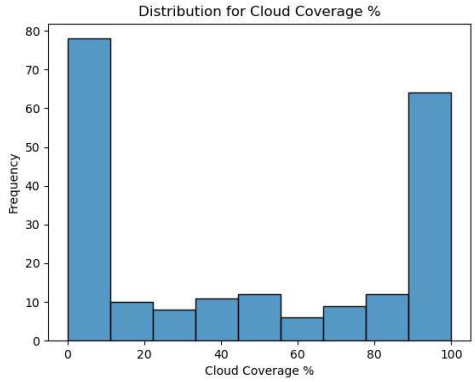


Here, we can notice that the training set has a small amount of values between 30-35, and then a higher number of values which are 35+, meaning our maximum will be around 35 and the outliers will be anything that's over 35. In the test set, however, there is a clear gap between 30-35, which will make our maximum threshold slightly lower (around the 30 degree mark) and making everything above that an outlier. Also, in the training set, due to an overwhelming majority of values being in the 15-25 range, and a pretty decent amount of values in the 0-5 range, our minimum will be in the middle of the 0-5 range, with anything lesser than this being considered an outlier. The distributions and boxplots are similar to the temperature graphs shown earlier due to the two sections having a high correlation.

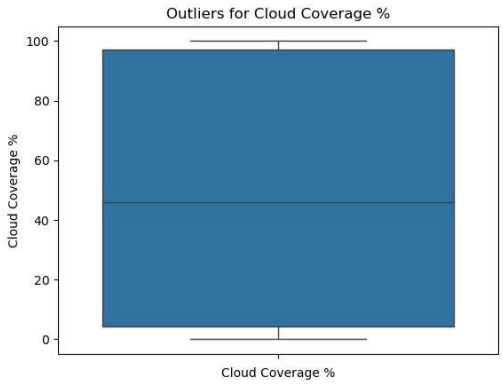


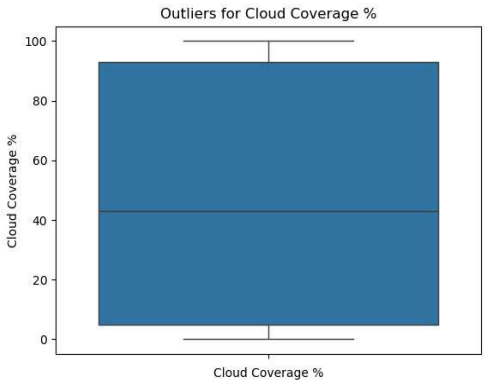


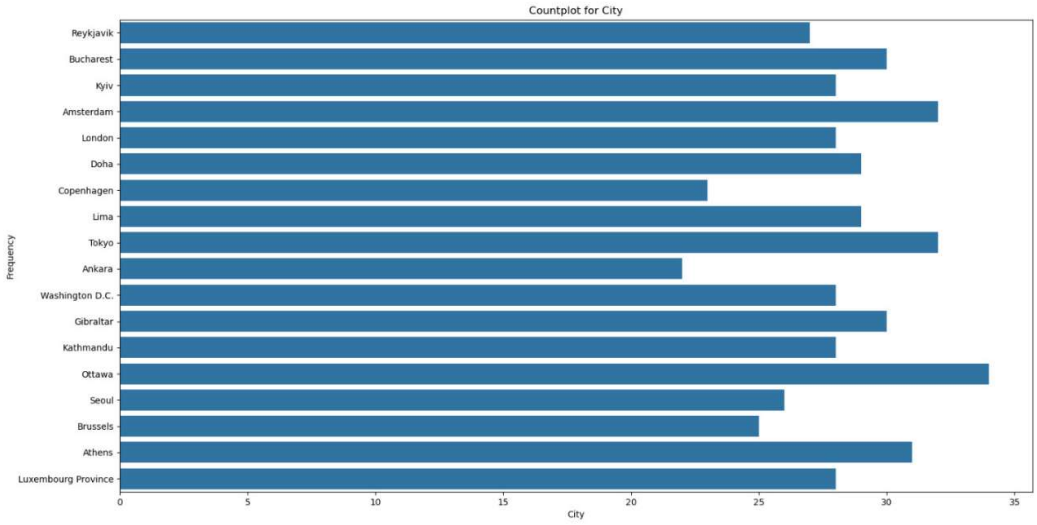


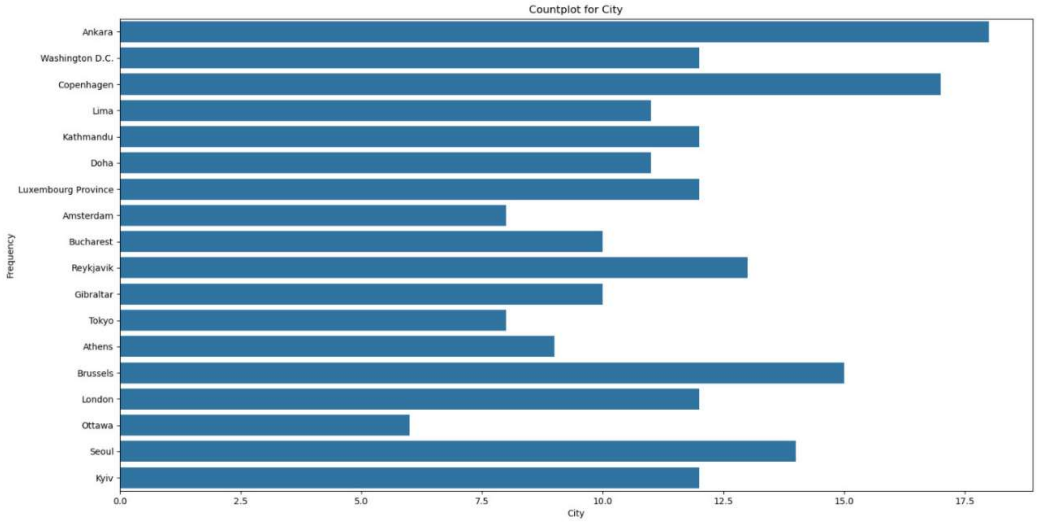


In the case of cloud coverage, due to both sets having a majority of values in either the low or high percentile, the boxplots will both be very spread out









Now, here come the heatmaps recorded in both sets:





Some things to note:

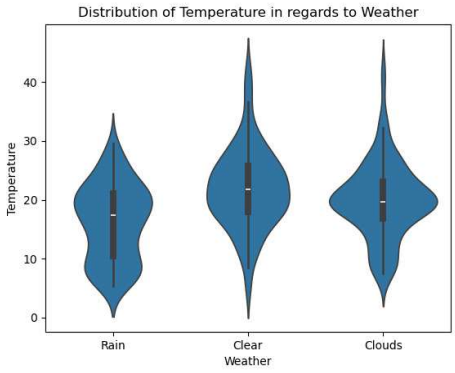
The real temperatures and the “feels like” temperatures obviously have a strong correlation

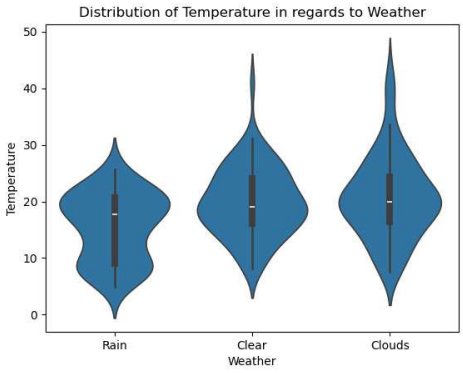
Wind speed and temperatures have close to no correlation

Humidity has a decent correlation with cloud coverage, and a strong inverse correlation with Temperature

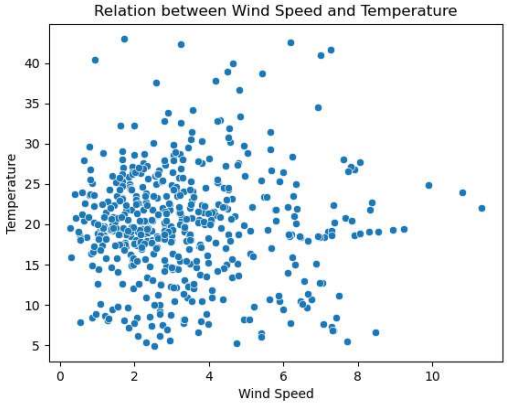
Cloud coverage has a small inverse correlation with temperature and a small inverse Correlation with wind speed

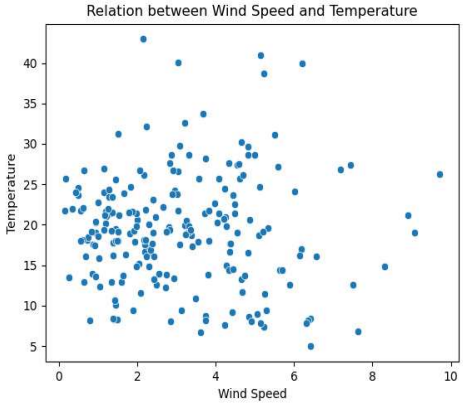
Now, I’ve chosen to analyze the distribution of temperatures in correlation with other characteristics by using violin and scatterplots. Here are the results:



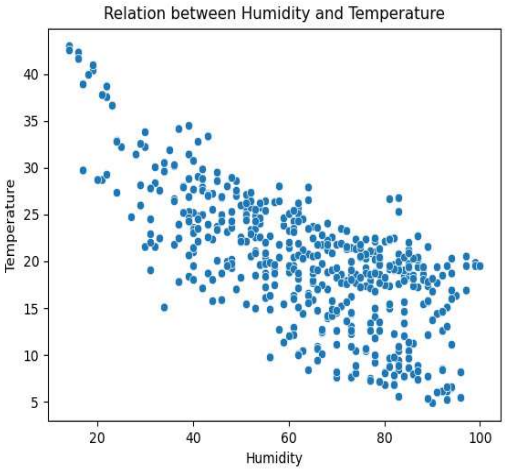


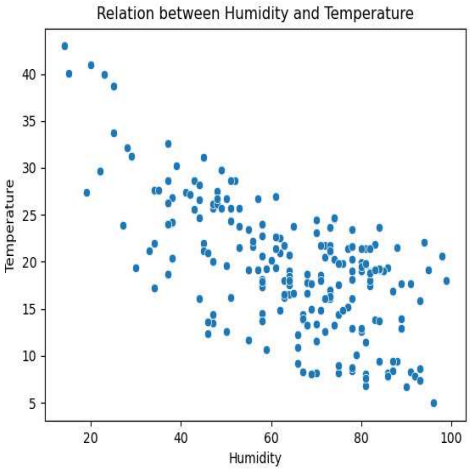
On average, the highest temperatures are when it's not raining, but clear weather has a more even distribution of medium temperatures than all the others, hence why it's got a smoother shape



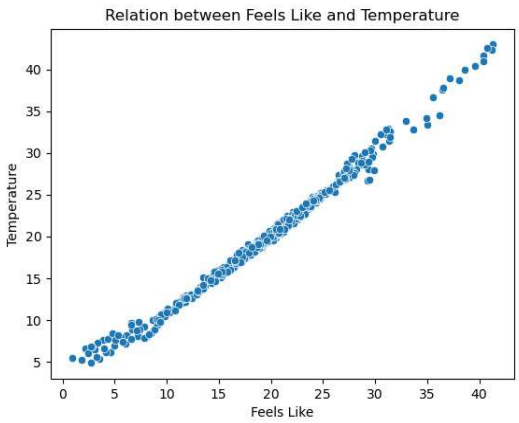


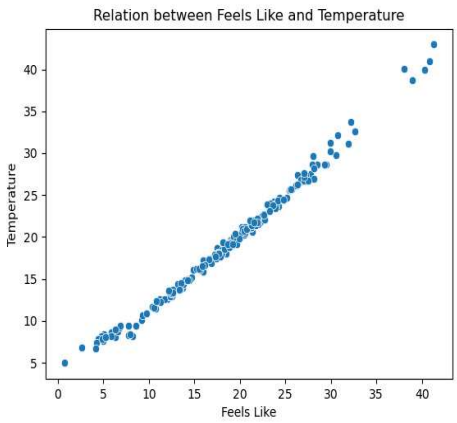
As I've mentioned previously, temperatures and wind speed have close to no correlation, hence why the values are scattered everywhere at random



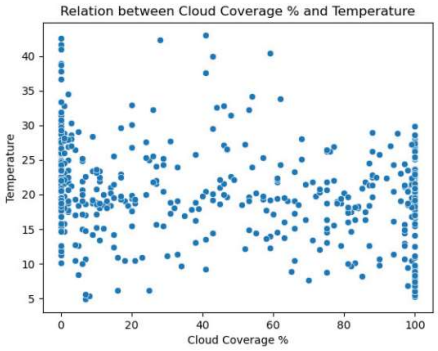


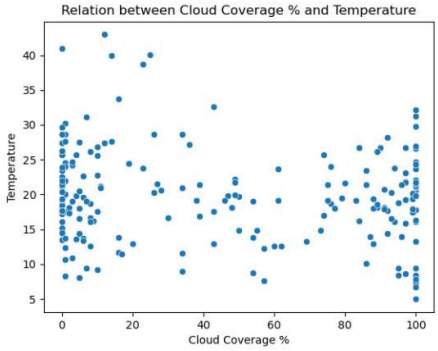
Humidity and temperature have a strong inverse correlation, hence why the values go from the upper right to the lower left, but it's not strong enough to group the values up nicely



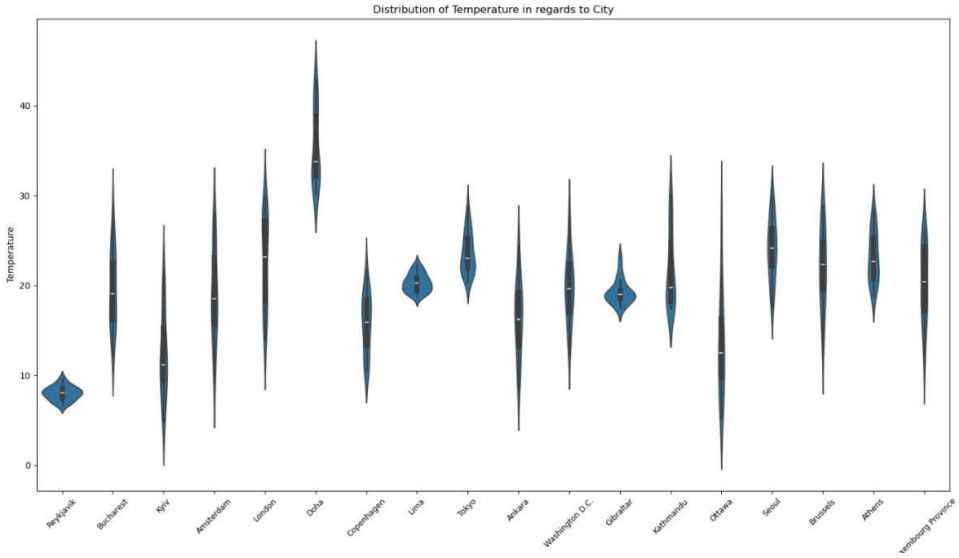


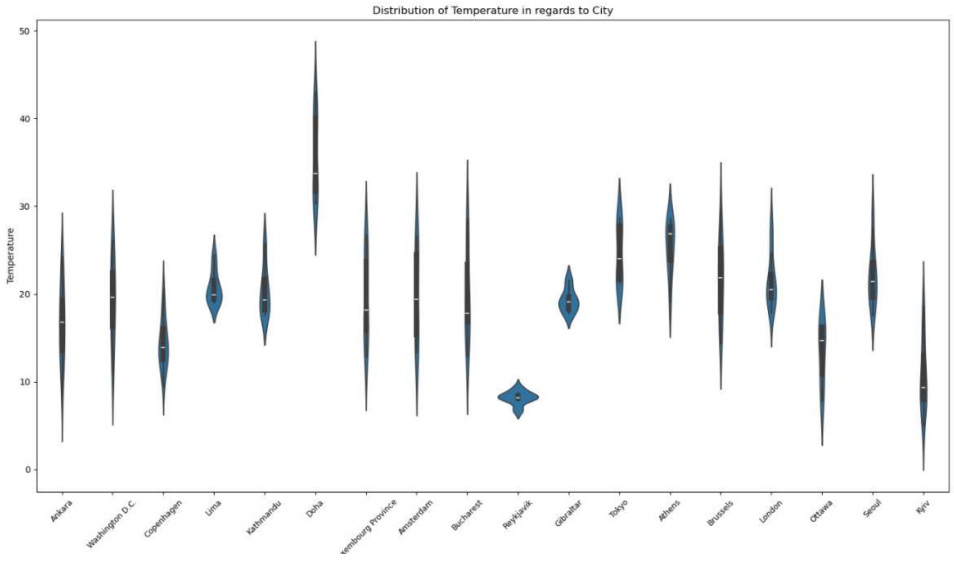
Obviously, the real temperatures and the “feels like” temperatures have a strong correlation, so they make a nice diagonal line; the reason for the gaps in both graphs is because both temperature distributions don’t have many temperatures in the 35-40 range





Cloud coverage and temperature have a small inverse correlation, so the values are all over the place





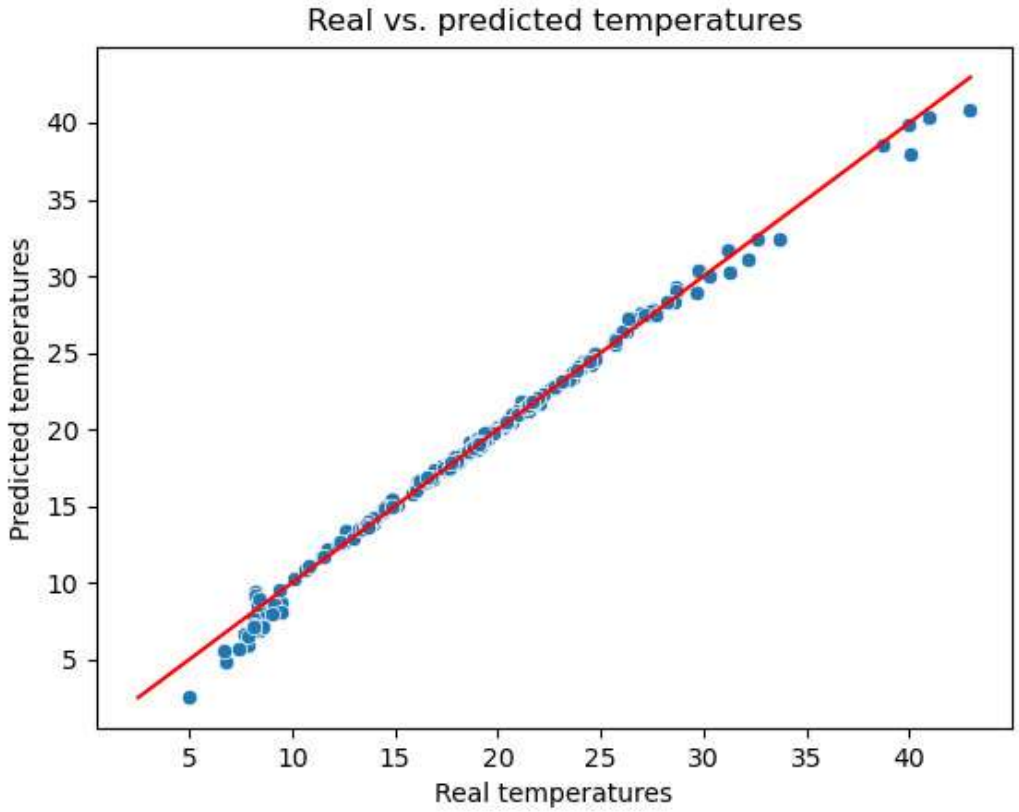
Here are the correlations between temperatures and cities in the form of a violinplot. For simplicity's sake, think of them as the range of temperatures in each capital city

Lastly, I chose to make a model that predicts temperatures by looking at wind speeds, humidity, cloud coverage percentages, and the "feels like" temperatures (which are a pretty big hint), by using linear regression.

It was trained using the training and testing sets. The errors are as such:

```
0.3266139610476788
0.5376331856204969
0.9938460890992623
```

The first error is the mean absolute error, which shows us that on average, temperatures are off by approximately 0.33 degrees C off, which is pretty decent. The second error is the root mean squared error, which isn't much higher than the MAE, implying that the model doesn't make any huge mistakes along the way. The last is the R squared score, which is *very* close to 1, telling us that it made almost perfect predictions. This is what the performance graph looks like:

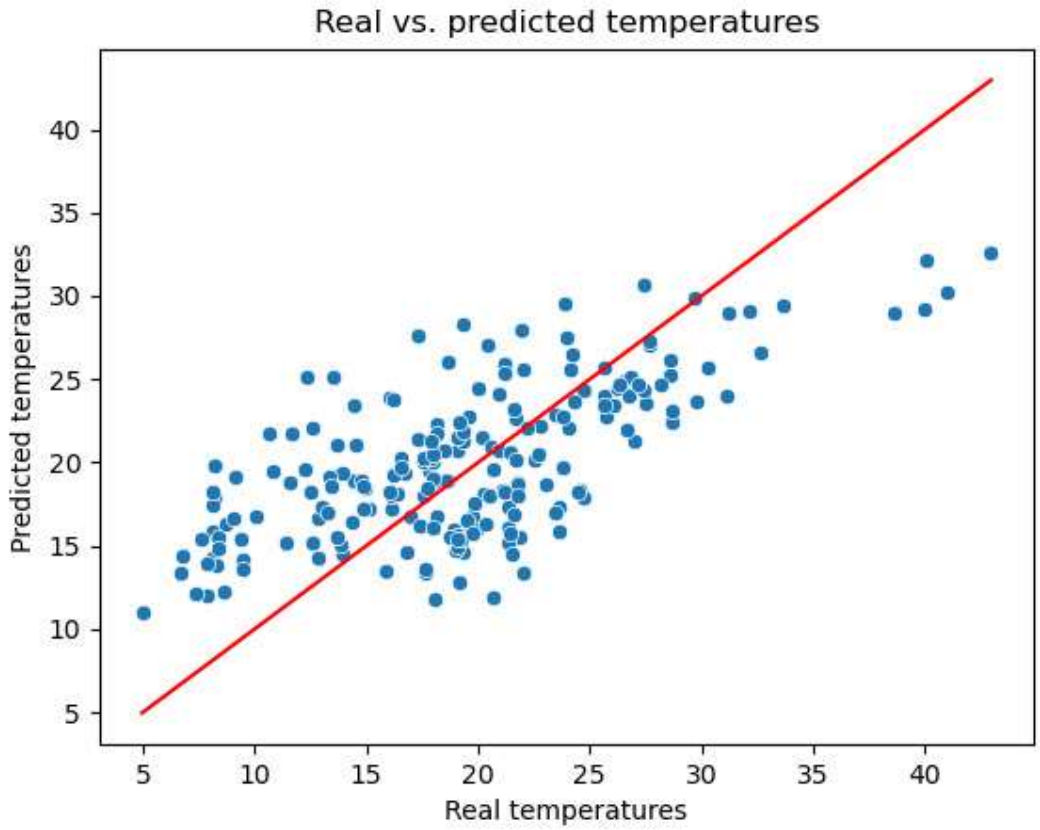


For starters, any value that is directly on the red line represents a perfect prediction. As we can see, in the areas where, in the training set, a majority of temperatures reside, the model made really good predictions, whereas on both ends of the line, where we have a minimal amount of temperatures in the training set, the model was pretty off because it didn't have much to train on. All in all, it gets an A for effort.

For scientific and comparison purposes, I decided to try and see how it would fare if it didn't have any "feels llike" temperatures. Due to all of the other characteristics having either close to zero correlation or strong negative correlations, it has performed pretty bad compared to what I've shown before. To begin, let's take a look at the errors:

```
4.273695537411125
5.116309644045882
0.4426952952848553
```

The MAE is now pretty big, being off by around 4.27 degrees C on average. The RMSE is now higher than the MAE by approximately 0.9, which is a really big difference compared to how it was before. And, sadly, the R squared score is now approximately 0.44, which is closer to 0 than it is to 1, meaning the model did a pretty poor job given the limited training set. Here's the performance graph:



As we can see, the values are now scattered all over the place due to the low correlation between the characteristics. For every point that's on the line, there are a few dozen which are way off.

In conclusion, giving a model a training set with characteristics that have a strong correlation with the target characteristic is crucial to getting good predictions. The stronger the correlations, the better the predictions, and vice versa.

This was all I've had to say about the project, it was pretty fun to do.

Thanks for reading and have a nice day!